

Amplitude cost of slope-constrained moment design: sharp asymptotics and a validated theta example

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Abstract

We minimize the amplitude of a bounded multiplier subject to finitely many exact weighted moments and a separate Lipschitz bound M . Under explicit simple-switch, separation, summability and moment-rank hypotheses, we prove $\delta(M) = \delta_* + C_* M^{-2} + o(M^{-2})$. The coefficient is $C_* = \delta_*^3 (3D_*)^{-1} \sum_k w(z_k) |r'_*(z_k)|$, where r_* is an optimal weighted L^1 residual, $D_* = \int w |r_*| du$, and z_k are its switches. The upper construction corrects all prescribed moments exactly by moving finitely many transition centers; the lower bound applies to every admissible multiplier. For a positive theta kernel at a certified triple zero, validated arithmetic verifies the hypotheses with infinitely many switches and encloses C_* in $(9.20340371, 9.20340375) \times 10^{-25}$. For every $M \geq 2 \times 10^{-5}$, the remainder lies between $-9.624 \times 10^{-33}/M^3$ and $2.167 \times 10^{-32}/M^3$. We also explain the relation to the classical moment duality and the two-parameter Kantorovich–Rubinstein norm. The numerical statements use analytic bounds and a companion computation archive; they have not been formally verified or externally refereed.

Keywords: moment problem; amplitude minimization; Lipschitz constraint; weighted L^1 approximation; validated numerics; positive kernels.

1 Problem and context

A bounded moment design can have an optimal multiplier that takes just two values and jumps between them. Requiring a finite slope replaces those jumps by transitions and increases the minimum amplitude. Our question is quantitative: how much amplitude must be added when the moments must still hold exactly? In an oscillatory integral, these moments can pin a zero and increase its multiplicity while the kernel remains positive. In the example below, an exact triple zero Q of a theta transform is held fixed. Three homogeneous moments preserve its first three vanishing derivatives, and a fourth moment cancels its nonzero third derivative. The optimization quantifies the smallest relative kernel change needed for this operation under a prescribed slope bound.

The setting is the half-line in its given coordinate u . For a positive density w , let $d\rho = w(u) du$, and let real functions $p_0, \dots, p_m$ belong to $L^1(\rho)$. Fix $f > 0$ and put

$$L_j(h) = \int_0^\infty p_j h d\rho, \quad \mathcal{A} = \{h : L_j(h) = 0 \ (j < m), \ L_m(h) = -f\}, \quad (1.1)$$

$$\delta_* = \inf_{h \in \mathcal{A} \cap L^\infty} \|h\|_\infty, \quad \delta(M) = \inf_{\substack{h \in \mathcal{A} \cap \text{Lip}([0, \infty)) \\ \text{Lip}(h) \leq M}} \|h\|_\infty. \quad (1.2)$$

Only bounded Lipschitz functions contribute a finite value in the second infimum. There is no constraint on the total mass of the kernel. The word *cost* means the relative amplitude $\|h\|_\infty$; it does not mean computation time or a physical resource.

The main result is Theorem 3.1, with a matching lower and upper coefficient. Section 4 verifies its hypotheses for a specific theta kernel. Theorem 5.1 supplies an explicit bound valid for every sufficiently large slope budget, with a specified threshold. These results do not characterize the exact finite- M minimizer.

Relation to established work. The L^1 approximation/ L^∞ moment duality is classical; a formulation of the Markov L -moment problem appears in Yuditskii [1, Problems 1.2–1.3]. Perfect-spline extremal problems relate moments to derivative norms under Chebyshev-system assumptions [2, Sections 7–8]. The oscillatory functions used here do not form a global Chebyshev system on the half-line. Amplitude and control-rate constraints also occur in minimum-time design [3]; hence the idea of imposing a hard slope bound is not new. Bang-bang regularization uses related information about small switching-function values [4], but a quadratic penalty parameter cannot be substituted for our hard bound M .

Our auxiliary support functional is exactly the two-parameter Kantorovich–Rubinstein (KR) norm of Lellmann et al. [5]. Its flat-metric and Wasserstein-type context is developed in [6, 7]. With the corresponding metric scaling it is also the Fortet–Mourier dual norm: the test-function norm uses a maximum of the amplitude and Lipschitz bounds, not their sum. Hille and Theewis give finite-dimensional reductions when one measure is atomic [8, Theorem 2.1] and norming sets of extreme test functions [9, Theorem 5.1 and Corollary 5.1]. These results describe the support functional; the exact-moment large- M coefficient below requires an additional argument. Section 2 records the precise equivalence, rather than presenting a known norm as a new one. The specific result studied here is the full coefficient with exact moment preservation, together with an infinite-switch validated example and a uniform remainder. The comparison concerns the cited results; it is not a claim that no equivalent theorem exists elsewhere in the literature.

2 Duality and normalization

For $a \in \mathbb{R}^m$, set

$$r_a = p_m - \sum_{j < m} a_j p_j, \quad D(a) = \int |r_a| d\rho. \quad (2.1)$$

Suppose that a_* satisfies

$$\int p_j \operatorname{sgn}(r_{a_*}) d\rho = 0 \quad (j < m), \quad D_* = D(a_*) > 0. \quad (2.2)$$

Write $r_* = r_{a_*}$. Then convexity shows that a_* minimizes D . Moreover

$$\delta_* = \frac{f}{D_*}, \quad h_* = -\delta_* \operatorname{sgn} r_*. \quad (2.3)$$

Indeed $\int p_m \operatorname{sgn} r_* d\rho = D_*$, so h_* is feasible. For every feasible h , $f = |\int r_* h d\rho| \leq \|h\|_\infty D_*$. If $r_* \neq 0$ almost everywhere, equality forces $h = h_*$ almost everywhere.

For completeness, write $q_j = w p_j$ and define

$$H(A, M; \mu) = \sup_{\|g\|_\infty \leq A, \operatorname{Lip}(g) \leq M} \int g d\mu, \quad (2.4)$$

$$V(A, M) = \sup_{\substack{\|g\|_\infty \leq A, \operatorname{Lip}(g) \leq M \\ \int q_j g = 0 \quad (j < m)}} \int q_m g. \quad (2.5)$$

For $A, M > 0$, the exact identities are

$$\begin{aligned} V(A, M) &= \inf_{a \in \mathbb{R}^m} H\left(A, M; (q_m - \sum_{j < m} a_j q_j) du\right), \\ \delta(M) &= \inf\{A > 0 : V(A, M) \geq f\}. \end{aligned} \quad (2.6)$$

To justify minimax, the set $\{|g| \leq A, \text{Lip}(g) \leq M\}$ is compact and convex in the topology of uniform convergence on compact intervals by diagonal Arzelà–Ascoli. Its moment maps are continuous: local convergence handles a compact interval and $A \int_U^\infty |q_j|$ controls the tails uniformly. Apply Sion’s theorem [10] to the affine Lagrangian. The infimum over the multipliers a imposes the lower moments exactly. The second identity follows by scaling a feasible maximizer of V by f/V whenever $V \geq f$.

Our convention is the *two separate bounds* in (2.4), with

$$H(A, M; \mu) = AH(1, M/A; \mu) = MH(A/M, 1; \mu). \quad (2.7)$$

Total variation, if used, is $\int |d\mu|$ without a factor $1/2$. Zero-extending a measure to the full line gives the same support functional: a half-line test function extends constantly to the left. Neither (2.6) nor our proof assumes that the unrestricted coefficient a_* remains dual-optimal for finite M .

3 Sharp large-slope asymptotics

Here are the complete local hypotheses used for the asymptotic result. Assume $m \geq 1$, $w > 0$, and $w, p_0, \dots, p_m$ are C^2 on the half-line (one-sided at zero), with $p_j \in L^1(\rho)$. Assume (2.2). In addition:

- (H1) $r_*(0) \neq 0$, and all positive zeros z_k of r_* are simple. The zero set is nonempty, finite or countably infinite. The signed set $\{z_k, -z_k\}$ is uniformly separated by a positive number d .
- (H2) For some fixed $b > 0$, the intervals $[z_k - b, z_k + b]$ are disjoint and contained in $(0, \infty)$, and for $0 \leq j \leq m$, $0 \leq \ell \leq 2$,

$$\sum_k \sup_{|u - z_k| \leq b} |q_j^{(\ell)}(u)| < \infty. \quad (3.1)$$

- (H3) There are m selected switches such that the matrix $(p_j(z_k))_{0 \leq j < m, 1 \leq k \leq m}$ is nonsingular, after relabeling.

In particular

$$0 < \Gamma = \sum_k w(z_k) |r'_*(z_k)| < \infty. \quad (3.2)$$

Finiteness follows from (3.1), since $(wr_*)'(z_k) = w(z_k)r'_*(z_k)$.

Theorem 3.1. *Under (H1)–(H3) and (2.2), the feasible Lipschitz class is nonempty for all sufficiently large M , and*

$$\delta(M) = \delta_* + \frac{C_*}{M^2} + o(M^{-2}), \quad C_* = \frac{\delta_*^3 \Gamma}{3D_*} = \frac{\delta_*^4 \Gamma}{3f} > 0. \quad (3.3)$$

If $\delta_ < 1$, the upper construction has $1 + h > 0$ for all sufficiently large M . The result is conditional on the stated switch and rank hypotheses.*

3.1 Lower bound for all feasible multipliers

Let h be feasible, $\text{Lip}(h) \leq M$, and $\delta = \|h\|_\infty$. The exact duality gap is

$$(\delta - \delta_*)D_* = \int_0^\infty |r_*|[\delta + \text{sgn}(r_*)h] d\rho. \quad (3.4)$$

The integrand is nonnegative. Fix finitely many switches. At z_k , put $\gamma_k = w(z_k)|r'_*(z_k)|$. In a sufficiently small neighborhood, $|(wr_*)(z_k \pm v)| \geq (\gamma_k - \varepsilon)v$ and the signs are opposite. The two bracketed defects $b_\pm$ in (3.4) satisfy

$$b_+ + b_- \geq 2\delta - 2Mv \geq 2\delta_* - 2Mv. \quad (3.5)$$

For sufficiently large M integrate over $0 \leq v \leq \delta_*/M$ to obtain

$$(\delta - \delta_*)D_* \geq \sum_{k \in E} (\gamma_k - \varepsilon) \int_0^{\delta_*/M} v(2\delta_* - 2Mv) dv = \frac{\delta_*^3}{3M^2} \sum_{k \in E} (\gamma_k - \varepsilon). \quad (3.6)$$

Taking the infimum over h , then the lower limit as $M \rightarrow \infty$, then $\varepsilon \downarrow 0$, and finally increasing the finite set E , proves $\liminf M^2(\delta(M) - \delta_*) \geq C_*$. This order of limits avoids a uniform Taylor assertion over infinitely many switches.

3.2 Exact moment correction and the matching upper bound

Evenly extend the sign profile $s = \text{sgn } r_*$ to the real line. Move only the selected positive centers to c_k , reflect them, and denote the new step profile by s_c . For $a > 0$ small set

$$s_{c,a}(u) = \frac{1}{2a} \int_{-a}^a s_c(u-v) dv, \quad S_j(c, a) = \int_0^\infty q_j(u) s_{c,a}(u) du. \quad (3.7)$$

Transitions are disjoint and avoid zero. Thus $\|s_{c,a}\|_\infty = 1$ and $\text{Lip}(s_{c,a}) = 1/a$. Negative reflected centers add no transition on the positive half-line. The jump at a positive switch is $d_k = 2 \text{sgn } r'_*(z_k)$.

For an upward unit step at c , its change in a moment with density q is

$$I_q(c, a) = -\frac{a}{2} \int_0^1 (1-v)[q(c+av) - q(c-av)] dv = -\frac{a^2}{6} q'(c) + o(a^2). \quad (3.8)$$

The same formula defines an even function of signed a at zero. If Q_ℓ bounds $|q^{(\ell)}|$ on the neighborhood, direct differentiation gives

$$|I_q| \leq a^2 Q_1/6, \quad |\partial_a I_q| \leq |a| Q_1/3, \quad |\partial_a^2 I_q| \leq Q_1/3 + |a| Q_2/12, \quad \partial_a^2 I_q(c, 0) = -q'(c)/3. \quad (3.9)$$

Hypothesis (H2) therefore permits summation and two differentiations of the local smoothing errors. Only finitely many centers move, so their center and mixed derivatives are local derivatives. These are sums of *local differences*, not sums of individual step tails. Consequently $S_{0:m-1}(c, a)$ has an even C^2 extension near $(z, 0)$.

At $(z, 0)$ its center Jacobian is

$$J_{jk} = -d_k q_j(z_k), \quad 0 \leq j < m, \quad 1 \leq k \leq m. \quad (3.10)$$

It is nonsingular by (H3). The implicit-function theorem supplies an even C^2 function $c(a) = z + O(a^2)$ with $S_j(c(a), a) = 0$ for every $j < m$.

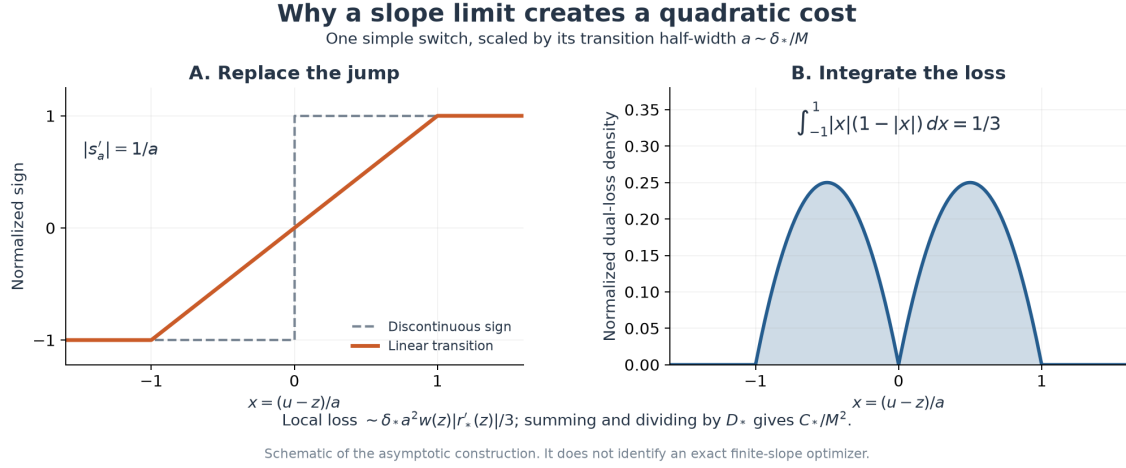


Figure 1: The local mechanism behind the coefficient $1/3$. A sign profile taking values ± 1 is replaced locally by a linear transition. The integral of $|x|(1 - |x|)$ on $[-1, 1]$ is $1/3$. The figure illustrates the construction; exact moment correction and the lower bound are supplied by the proof.

Let $q_r = wr_*$. Since $q_r(z_k) = 0$, moving finitely many centers by $O(a^2)$ changes its unsmoothed moment by $O(a^4)$. The corresponding change in the smoothing correction is also $O(a^4)$, by the local C^2 bounds. Summing (3.8) with the factors d_k yields

$$S_m(c(a), a) = \int q_r s_{c(a),a} = D_* - \frac{\Gamma a^2}{3} + o(a^2). \quad (3.11)$$

Define

$$\alpha(a) = \frac{f}{S_m(c(a), a)}, \quad h_a = -\alpha(a) s_{c(a),a}. \quad (3.12)$$

All moments in (1.1) are exact. Furthermore

$$\|h_a\|_\infty = \alpha(a) = \delta_* + \frac{\delta_* \Gamma}{3D_*} a^2 + o(a^2), \quad \text{Lip}(h_a) = \frac{\alpha(a)}{a}, \quad \alpha'(a) = O(a). \quad (3.13)$$

The map $a \mapsto \alpha(a)/a$ is strictly decreasing for small positive a and tends to infinity. Thus every sufficiently large M is obtained, with $a(M) = \delta_*/M + O(M^{-3})$. Substitution gives the matching upper bound in (3.3), completing the proof. $\square$

4 A positive theta kernel at a pinned triple zero

Consider the fixed-kernel family

$$\Phi(u) = \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \quad (4.1)$$

$$F_h(t, \lambda, \mu, \nu) = \int_0^\infty \Phi(u) (1 + h(u)) e^{\lambda u^2 + \mu u^4 + \nu u^6} \cos(2tu) du. \quad (4.2)$$

Every summand of Φ is positive for $u \geq 0$. Double exponential decay gives joint analyticity in the four moving coordinates for bounded h . The multiplier h is a single fixed function, independent of those coordinates.

Let $Q = (\tau, \lambda_Q, \mu_Q, 0)$ be the unique solution of $F_0 = (F_0)_t = (F_0)_{tt} = 0$ in the exact box of certificate C1 (Appendix A). Its approximate location is

$$(\tau, \lambda_Q, \mu_Q) \simeq (41.40034135868425, -3.645692061604919, 8.335120498918607). \quad (4.3)$$

These decimals locate Q ; they do not replace the exact dyadic center and the certified displacement bound. Set

$$w = \Phi e^{\lambda_Q u^2 + \mu_Q u^4}, \quad p_j = (2u)^j \cos(2\tau u + j\pi/2), \quad f_j = \int p_j d\rho. \quad (4.4)$$

Here $f_0 = f_1 = f_2 = 0$ and $f = f_3 > 0$. Taking $m = 3$ in (1.1) pins Q and makes $\partial_t^j F_h(Q) = 0$ for $0 \leq j \leq 3$. It imposes a zero of order *at least* four.

4.1 The geometric interpretation

The parameter identities are

$$(F_h)_\lambda = -\frac{1}{4}(F_h)_{tt}, \quad (F_h)_\mu = \frac{1}{16}(F_h)_{t^4}, \quad (F_h)_\nu = -\frac{1}{64}(F_h)_{t^6}. \quad (4.5)$$

At the unmodified Q , certificate C1 encloses

$$\begin{aligned} 3.33563055 \times 10^{-13} &< f_3 < 3.33563057 \times 10^{-13}, \\ -9.71679534 \times 10^{-13} &< f_4 < -9.71679532 \times 10^{-13}. \end{aligned} \quad (4.6)$$

The two-control determinant is $f_4 f_3 / 64 \neq 0$. Thus Q is a transverse triple zero, called a cusp of the equilibrium equation $F_0 = 0$. For a local potential with $V_t = F_0$, this is the A_3 convention.

Write $g_j = \partial_t^j F_h(Q)$. At the constrained target, direct use of (4.5) gives

$$\det \partial_{(\lambda, \mu, \nu)}(g_0, g_1, g_2) = \frac{g_4(g_4 g_7 - g_5 g_6)}{4096}. \quad (4.7)$$

A nonzero value ensures both exact order four and rank three of the unfolding. This is the swallowtail (A_4) condition in the potential convention, not a statement about a fourth-order zero of the original unmodified transform.

Proposition 4.1. *For this theta example and every $M > 0$, $\delta(M)$ is also the infimum over smooth multipliers with $\text{Lip}(h) \leq M$, $\|h\|_\infty < 1$, the same four exact moments, and nonzero determinant (4.7). This is equality of infima, not attainment in the smooth class.*

Proof. The functions $p_0, \dots, p_N$ are independent on every open interval. A relation would give $P(u) \cos(2\tau u) + Q(u) \sin(2\tau u) = 0$ with polynomials P, Q . Analytic continuation would express $e^{4i\tau z}$ as a rational function unless $P = Q = 0$, which is impossible since $\tau \neq 0$. For a nonnegative compactly supported smooth bump χ positive on an interval, the Gram matrix $B_{ij} = \int \chi p_i p_j d\rho$ is positive definite. Hence

$$R_N v = \chi \sum_{j=0}^N (B^{-1} v)_j p_j \quad (4.8)$$

is a right inverse of the first $N + 1$ moments, bounded in the C^1 norm.

Since $\delta_* < 1$ (Table 1), first cut off h_* and then mollify its even extension using nonnegative kernels. The resulting smooth compactly supported functions have amplitude at most δ_* and converge in the weighted moments through order seven. The strict gap between δ_* and one

Quantity	Certified enclosure or bound
Unrestricted amplitude δ_*	$(9.1787079603827, 9.1787079608363) \times 10^{-10}$
Weighted switch sum Γ	$(1.297542117, 1.297542121)$
Leading coefficient C_*	$(9.20340371, 9.20340375) \times 10^{-25}$
Positive switches in $[0, 1]$	28 simple roots; 113 complementary sign leaves
Tail of Γ beyond $u = 1$	$< 5.705 \times 10^{-63}$
Signed-switch separation	$> 1/200$
First three switch moment determinant	positive, approximately 5.30861×10^{-6}

Table 1: Outward decimal bounds from certificates C2–C3. The last approximation is a locator; exclusion of zero is checked with the exact interval in C3. No rounded center is treated as an exact optimizer.

leaves room for the following corrections. Use R_3 to correct the first four moment errors exactly, and an arbitrarily small R_7 correction in jets 4–7 to avoid the zero set of the nonzero polynomial in (4.7). This gives a smooth feasible anchor h_0 with $\|h_0\|_\infty < 1$ and finite slope. The family $h_\theta = -1 + \theta(1 + h_0)$, $0 < \theta \leq 1$, remains feasible, has $1 + h_\theta > 0$, and has arbitrarily small slope. Its nonzero jets and determinant scale by θ and θ^3 , respectively.

For any $M > 0$ a bounded feasible anchor exists. A minimizing sequence has a locally uniformly convergent subsequence by Arzelà–Ascoli; integrable tails preserve the moments. Thus a Lipschitz minimizer exists, and the anchor shows $\delta(M) < 1$. Mix it with a smooth anchor of slope strictly less than M . The mixture has a strict slope margin and cost arbitrarily close to $\delta(M)$. Even extension followed by a nonnegative even mollifier preserves the amplitude and does not increase the slope. Moment errors tend to zero. Correct them with R_3 ; then perturb jets 4–7 with R_7 to make (4.7) nonzero. Both C^1 corrections can fit the strict slope and positivity margins. This proves the claim. $\square$

4.2 Validated hypotheses and the leading coefficient

The weighted L^1 objective $D(a)$ is coercive: independence of the first three p_j gives $\|\sum a_j p_j\|_{L^1(\rho)} \geq c|a|$. Every residual is nonzero analytic, so its zeros have measure zero and D is differentiable, with gradient given by the sign moments. The following bounds apply to the *exact* Q .

Certificate C3 starts from an exact dyadic vector $\bar{a}$, approximately

$$\bar{a} \simeq (-0.0001172686588251, -0.0646797594922349, 0.00388087037678106). \quad (4.9)$$

Throughout $\|a - \bar{a}\|_\infty \leq R = 2^{-40}$, it isolates the 28 simple roots in $[0, 1]$ and excludes other roots there by a sign cover. Let I_k be fixed disjoint neighborhoods containing these root boxes, with no other zero of any residual in the coefficient cube. If w_1 is the first positive theta summand times the deformation factor, define

$$D_{\text{loc}}(a) = \int_{\bigcup_{k=1}^{28} I_k} w_1(u) |r_a(u)| \, du. \quad (4.10)$$

Its Hessian is

$$H(a) = 2 \sum_{k=1}^{28} \frac{w_1(z_k(a))}{|r'_a(z_k(a))|} p(z_k(a)) p(z_k(a))^T, \quad p = (p_0, p_1, p_2)^T, \quad (4.11)$$

Positive leading principal minors certify $H - (1/200)I \succ 0$ uniformly. The difference $D - D_{\text{loc}}$ is an integral of $|r_a|$ against a nonnegative density on a fixed set, hence is convex. Therefore

D is strongly convex on the cube. Also $\|\nabla D(\bar{a})\|_1 < 2.065 \times 10^{-16}$ and $R/200 - 2\|\nabla D(\bar{a})\|_1 > 4.1345 \times 10^{-15}$. Strong convexity makes $D > D(\bar{a})$ on the Euclidean sphere of radius R , giving an interior stationary point. Convexity makes it a global minimizer; local strict convexity rules out any second global minimizer. Thus the exact a_* is unique and localized in the recorded cube.

For $u \geq 1$ the residual is

$$r_a = A(u) \sin(2\tau u) + B(u) \cos(2\tau u), \quad A = 8u^3 + 2a_1u, \quad B = 4a_2u^2 - a_0. \quad (4.12)$$

The uniform coefficient bounds $|a_0| < 10^{-3}$, $|a_1| < 1/10$, $|a_2| < 1/200$ imply $A > 7u^3$. With $\vartheta = \arctan(B/A)$,

$$AB' - BA' = -32a_2u^4 + (8a_1a_2 + 24a_0)u^2 + 2a_0a_1, \quad |\vartheta'| \leq (200u^2)^{-1}. \quad (4.13)$$

The phase derivative lies between 81 and 85. Consequently there are infinitely many simple tail switches, spaced by more than $3/85$. The finite cover and the transition across $u = 1$ complete (H1).

Differentiating the theta series gives, for $\ell = 0, 1, 2$,

$$|\Phi^{(\ell)}(u)| \leq C_\ell e^{(9+4\ell)u - \pi e^{4u}}, \quad \begin{cases} C_0 = 4\pi^2 + 6\pi, \\ C_1 = 60\pi^2 + 30\pi + 16\pi^3, \\ C_2 = 660\pi^2 + 150\pi + 448\pi^3 + 64\pi^4. \end{cases} \quad (4.14)$$

For example, the derivative polynomials are generated by $T(P) = (5 - 4x)P + 4xP'$ from $P_0 = -3 + 2x$. Their absolute monomial terms have successive-index ratios at most $256e^{-3\pi} < 1/2$; twice the first term bounds each series. Polynomial factors times this double exponential envelope prove (H2). The first three switches satisfy (H3). The finite contributions to Γ use 12 theta terms with an explicit series tail; beyond $u = 1$ one uses $|r'_a| < 900u^3$ and the phase spacing. In particular

$$\Gamma_{u>1} \leq \frac{900C_0 \exp(9 + \mu_+ - \pi e^4)}{1 - 50^{-4}} < 5.705 \times 10^{-63}, \quad (4.15)$$

where μ_+ is the recorded upper bound for μ_Q . For this envelope the logarithmic derivative is $3/u + 9 + 4\mu_+u^3 - 4\pi e^{4u}$. After division by e^{4u} , each positive term is decreasing for $u \geq 1$. Thus its value at one bounds the decay rate throughout the tail. That rate times $3/85$ exceeds 16 and $e^4 > 50$, giving the displayed geometric denominator. These checks give Table 1 and apply Theorem 3.1.

5 An explicit uniform remainder

Theorem 5.1. *For the exact theta problem in Section 4, for every $M \geq M_0 = 2 \times 10^{-5}$,*

$$\frac{C_*}{M^2} - \frac{9.624 \times 10^{-33}}{M^3} < \delta(M) - \delta_* < \frac{C_*}{M^2} + \frac{2.167 \times 10^{-32}}{M^3}. \quad (5.1)$$

In particular

$$\left| \frac{\delta(M) - \delta_*}{C_*/M^2} - 1 \right| < 0.001178 \frac{M_0}{M}. \quad (5.2)$$

No sharpness of the order M^{-3} or second asymptotic coefficient is asserted. The relative error is for the excess above δ_ .*

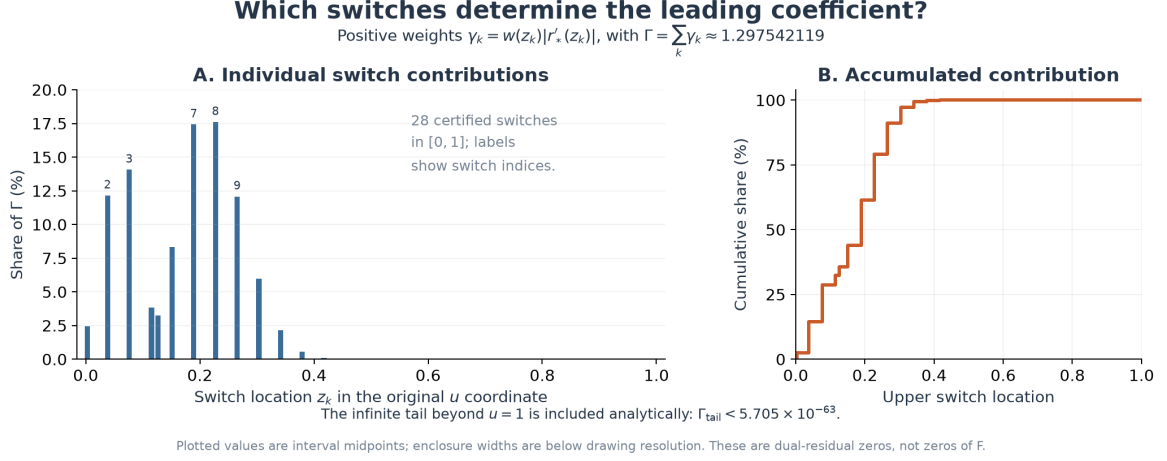


Figure 2: Contributions to Γ in the theta example. The plotted values are interval midpoints, with widths below the drawing resolution; the infinite tail is included analytically. These switches are zeros of the optimal dual residual, not zeros of the transform F_h .

5.1 Uniform correction over a continuum of transition widths

Put $a_{\max} = 2^{-14}$ and $V = (512, 768, 384)$. Move only the first three centers: $c_k = z_k + a^2 v_k$, $|v_k| \leq V_k$, $0 < a \leq a_{\max}$. Let U_k contain the exact root and a radius $a_{\max} + a_{\max}^2 V_k$ for $k \leq 3$, or $a_{\max}$ otherwise, and define

$$Q_{j\ell k} = \sup_{U_k} |q_j^{(\ell)}|, \quad R_{\ell k} = \sup_{U_k} |q_r^{(\ell)}|, \quad (5.3)$$

$$B_2 = \sum_k R_{2k}, \quad A_3 = B_2/12, \quad A_4 = \sum_{k=1}^3 (R_{1k} V_k^2 + R_{2k} V_k/3).$$

Both B_2 and A_3 include the infinite tail; A_4 includes only the three moved centers. Certificate C4 supplies

$$B_2 < 59.125014562, \quad A_3 < 4.927084547, \quad A_4 < 132801.204. \quad (5.4)$$

Taylor's theorem in (3.8) gives

$$|I_q(c, a) + a^2 q'(c)/6| \leq a^3 \sup |q''|/24. \quad (5.5)$$

Let $F_j = \frac{1}{6} \sum_k d_k q'_j(z_k)$ and let J be (3.10). Then $|S_j(z, a)/a^2 + F_j| \leq (a/12) \sum_k Q_{j2k}$. For the exact dyadic preconditioner B recorded in C4, use

$$T_a(v) = v - BS_{0:2}(z + a^2 v, a)/a^2, \quad \|v\|_V = \max_k |v_k|/V_k. \quad (5.6)$$

The exact center derivative is $-d_k$ times the symmetric average of q_j on $[c_k - a, c_k + a]$. Hence

$$|\partial_{c_k} S_j - J_{jk}| \leq \Delta_{jk} := 2a_{\max}^2 (V_k Q_{j1k} + Q_{j2k}/6). \quad (5.7)$$

With $E = I - BJ$, define the row bounds

$$\kappa_i = V_i^{-1} \sum_k (|E_{ik}| + \sum_j |B_{ij}| \Delta_{jk}) V_k, \quad (5.8)$$

$$\eta_i = V_i^{-1} \left(|(BF)_i| + \frac{a_{\max}}{12} \sum_j |B_{ij}| \sum_k Q_{j2k} \right). \quad (5.9)$$

Outward bounds are

$$\kappa < (0.129747, 0.118233, 0.071040), \quad \eta < (0.397772, 0.406040, 0.324017). \quad (5.10)$$

Thus T_a is a uniform contraction mapping the V -box to its interior. Since $\|I - BJ\|_V < 1$, BJ and therefore B are invertible; its fixed point solves $S_{0:2} = 0$. Also, uniformly in bounded v ,

$$S_{0:2}(z + a^2v, a)/a^2 = Jv - F + O(a). \quad (5.11)$$

Thus $T_0(v) = (I - BJ)v + BF$ is its continuous extension to $a = 0$. Uniform contraction gives continuous fixed points $v(a)$ for the entire interval. This is a continuum argument, not a sampling of budgets.

5.2 Upper and lower error bounds

Let $L(a) = D_* - \int q_r s_{c(a),a}$. Positivity of the sign deficit gives $L \geq 0$. Put $\Delta_k = c_k - z_k$. Since $q_r(z_k) = 0$, the unsmoothed center displacement satisfies

$$\left| \int q_r(s_c - s_z) du \right| \leq \sum_{k=1}^3 R_{1k} |\Delta_k|^2. \quad (5.12)$$

Indeed the jump has magnitude two and $|q_r(z_k + x)| \leq R_{1k}|x|$. Differentiating (3.8) in c also gives $|\partial_c I_{q_r}| \leq a^2 R_{2k}/6$, and hence

$$|d_k[I_{q_r}(c_k, a) - I_{q_r}(z_k, a)]| \leq a^2 R_{2k} |\Delta_k|/3. \quad (5.13)$$

Using $|\Delta_k| \leq a^2 V_k$ produces exactly the finite sum A_4 . At the original centers, summing (5.5) produces A_3 . Therefore

$$L(a) \leq \Gamma a^2/3 + A_3 a^3 + A_4 a^4. \quad (5.14)$$

Let f_- be the certified lower bound for f_3 , and let U_δ , Γ_+ and C_+ denote certified upper bounds for δ_* , Γ and C_* , respectively. Define

$$D_{\min} = f_-/U_\delta, \quad E_0 = \Gamma_+ a_{\max}^2/3 + A_3 a_{\max}^3 + A_4 a_{\max}^4, \quad (5.15)$$

$$\bar{U} = \frac{U_\delta}{1 - E_0/D_{\min}}. \quad (5.16)$$

Upper bounds are used on the right. C4 checks $E_0 < D_{\min}$, $\bar{U} < 1$, and $\bar{U}/a_{\max} < M_0$. For the feasible ramp $h_a = -\alpha(a)s_{c(a),a}$, $\alpha = f_3/(D_* - L)$ lies between δ_* and $\bar{U}$. Continuity, the divergence of $\alpha(a)/a$ at zero, and its value below M_0 at $a_{\max}$ give every $M \geq M_0$ by the intermediate-value theorem. Monotonicity on this full interval is not required.

Put $e_a = \alpha - \delta_*$ and choose $a = \alpha/M$. The exact relation is $D_* e_a = \alpha L$, so

$$D_* e_a \leq \frac{\Gamma \alpha^3}{3M^2} + \frac{A_3 \alpha^4}{M^3} + \frac{A_4 \alpha^5}{M^4}. \quad (5.17)$$

Using $\alpha^3 - \delta_*^3 \leq 3\bar{U}^2 e_a$ and absorbing this difference,

$$e_a - C_*/M^2 \leq \frac{A_3 \bar{U}^4/M^3 + (A_4 \bar{U}^5 + \Gamma \bar{U}^2 C_+)/M^4}{D_* - \Gamma \bar{U}^2/M^2}. \quad (5.18)$$

The denominator is positive. Exact rational arithmetic on the certified enclosures gives

$$\frac{A_3 \bar{U}^4 + (A_4 \bar{U}^5 + \Gamma_+ \bar{U}^2 C_+)/M_0}{D_{\min} - \Gamma_+ \bar{U}^2/M_0^2} < 2.167 \times 10^{-32}. \quad (5.19)$$

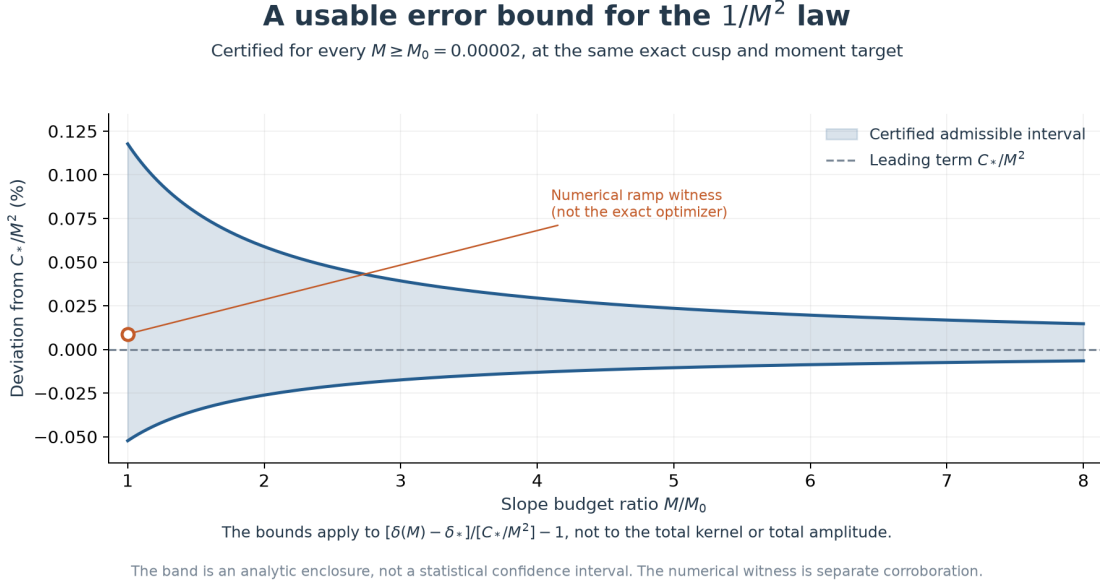


Figure 3: The enclosure for the deviation from the leading excess C_*/M^2 . The band follows from (5.1); it is not a statistical confidence interval. The hollow point is a separate numerical ramp witness and does not assert an exact finite-budget optimizer.

Since $\delta(M) \leq \alpha$, this proves the upper inequality.

For any admissible h , Taylor’s theorem gives $|q_r(z_k \pm v)| \geq P_k(v) := \gamma_k v - R_{2k} v^2/2$. The true weights and the defects $b_{\pm}$ in the lower-bound proof are nonnegative. For $0 \leq v \leq \delta_*/M$ their paired contribution is at least

$$\max(0, P_k)(b_+ + b_-) \geq \max(0, P_k)(2\delta_* - 2Mv) \geq P_k(2\delta_* - 2Mv). \quad (5.20)$$

This positive-part step matters when the Taylor lower bound is negative. Integration gives $\gamma_k \delta_*^3/(3M^2) - R_{2k} \delta_*^4/(12M^3)$. Sum finite sets and pass to the absolutely convergent sums to obtain

$$\delta(M) - \delta_* \geq C_*/M^2 - \frac{\delta_*^5 B_2}{12f_3 M^3}, \quad \frac{U_\delta^5 B_2}{12f_-} < 9.624 \times 10^{-33}. \quad (5.21)$$

The neighborhoods contain the required intervals for all $M \geq M_0$. The strict slack in the rounded constants gives (5.1). Dividing by the lower bound on C_* proves (5.2). $\square$

At the threshold M_0 the total amplitude is enclosed by

$$\begin{aligned} 0.00000000091787309568619750 &< \delta(M_0) \\ &< 0.00000000091787309964331750. \end{aligned} \quad (5.22)$$

The bound in (5.2) is below 0.1178% there, for the much smaller *excess*, not the total amplitude. The theorem applies to the smooth nondegenerate class as an infimum by Proposition 4.1. Here “large slope” is relative to the amplitude: at M_0 the leading transition half-width δ_*/M_0 is approximately 4.5894×10^{-5} in the original u coordinate, well below the switch separation.

6 Interpretation and scope

The baseline amplitude δ_* measures the least bounded relative change that raises the multiplicity at a fixed triple zero. A finite slope adds a positive cost whose leading coefficient weights

every simple switch by the local density and the slope of the optimal residual. Even though there are infinitely many switches in this example, their contributions are summable and the tail is bounded explicitly. The scale and the coefficient depend on the original coordinate u ; rescaling that coordinate also rescales the Lipschitz budget.

The conditional theorem extends beyond the theta density, but it requires verification of its hypotheses for each new example. It does not cover multiple residual zeros, accumulating switches, a boundary switch, or a singular selected moment matrix. The validated application is at one exact base point Q , not a uniform result along the entire cusp curve. It gives no global count of transform zeros or swallowtails, no result about undeformed zeta zeros, and no physical model or trained statistical model. The feasible ramps used for the upper bound are not proved to be the exact finite- M optimizers, and smooth attainment is not claimed.

A Certificate interface and reproducibility

The companion research archive contains exact inputs, analytic bounds, producer programs, separate arithmetic checkers, numerical crosschecks, and file hashes. Table 2 defines the certificate identifiers used above. Paths are relative to the archive’s `research/` folder. The full filename in each folder is under `results/`.

ID	Folder	Certificate filename
C1	<code>cusp_verified</code>	<code>quartic_cusp_certificate.json</code>
C2	<code>kernel_norm_threshold</code>	<code>threshold_certificate.json</code>
C3	<code>kernel_slope_asymptotics</code>	<code>asymptotic_certificate.json</code>
C4	<code>kernel_slope_remainder</code>	<code>remainder_certificate.json</code>

Table 2: Machine-readable companions of the numerical claims. Their SHA-256 identities and the claim-to-source map are recorded in the manuscript package. The archive is presently a local research artifact; a public version and persistent identifier remain to be assigned.

Arithmetic representation and the base point. A recorded ball has midpoint $m2^e$ and radius $r2^s$, with exact integers. The enclosing set is $[m2^e - r2^s, m2^e + r2^s]$; printed decimal summaries are not inputs. C1 applies a fixed-preconditioner contraction to $G = (F_0, (F_0)_t, (F_0)_{tt})$ on a cube of radius approximately 10^{-24} . With a uniform Hessian bound M_2 , it verifies $q = \|I - YDG(x_0)\| + \|Y\|M_2R < 5.408 \times 10^{-13}$ and $\eta + qR < R$, where $\eta = \|YG(x_0)\|$. Nonsingularity of Y and Banach’s theorem give the unique zero and $\|Q - x_0\|_\infty \leq \eta/(1 - q) < 9.099 \times 10^{-88}$. Derivative uncertainty is propagated from the exact center to that root.

The finite theta sum is integrated on $[0, U]$ with validated complex quadrature using Arb [11, 12]; separate bounds cover the omitted theta terms and $u > U$. For C_0 in (4.14) and $P^+(u) = \lambda_+ u^2 + \mu_+ u^4 + \nu_+ u^6$, where each coefficient is the maximum of zero and its upper parameter-box bound, one may use

$$\int_U^\infty |\partial_t^n(\Phi e^P \cos(2tu))| du \leq \frac{C_0(2U)^n}{c_n} e^{9U+P^+(U)-\pi e^{4U}}, \quad (\text{A.1})$$

$$c_n = 4\pi e^{4U} - 9 - n/U - (P^+)'(U) > 0.$$

The choice of U also makes each positive polynomial derivative term times e^{-4u} nonincreasing, so the bound holds on the entire tail. C1 uses $U = 2$, 16 theta terms and 110 decimal working precision, with a separate higher-precision check. C2 obtains a lower amplitude bound from an upper bound for D , and an upper amplitude bound from an exactly moment-corrected feasible design. Thus the two certificate directions are not interchanged.

Remainder derivative tails. C4 uses (4.14) and $|P'| \leq 44(1+u)^3$, $|P''| \leq 116(1+u)^2$. Let

$$W_0 = C_0, \quad W_1 = C_1 + 44C_0, \quad W_2 = C_2 + 88C_1 + 2052C_0, \quad (\text{A.2})$$

$$T_{j0} = 2^j W_0, \quad T_{j1} = 2^j [W_1 + (j+84)W_0]. \quad (\text{A.3})$$

For $j \leq 3$ set $T_{j2} = 2^j [W_2 + 2(j+84)W_1 + (j(j-1) + 168j + 7056)W_0]$. Then, for $0 \leq \ell \leq 2$,

$$|q_j^{(\ell)}(u)| \leq T_{j\ell}(1+u)^9 \exp(17u + \mu_+ u^4 - \pi e^{4u}). \quad (\text{A.4})$$

For $u_0 = 1 - a_{\max}$ the logarithmic decay rate of this envelope is at least $c = 4\pi e^{4u_0} - 4\mu_+ u_0^3 - 17 - 9/(1+u_0)$ throughout $u \geq u_0$. To see why a bound at u_0 suffices, divide the logarithmic derivative $9/(1+u) + 17 + 4\mu_+ u^3 - 4\pi e^{4u}$ by e^{4u} . All three positive terms are decreasing for $u \geq u_0 > 3/4$. The logarithmic derivative is consequently at most $-ce^{4(u-u_0)} \leq -c$. Since $c(3/85) > 16$, the sum of neighborhood suprema is bounded by the envelope at u_0 divided by $1 - 50^{-4}$. Combining these bounds with the coefficient bounds gives the q_r tails. Finite interval evaluations and these infinite-tail bounds, rather than samples of derivatives, supply the constants in (5.3).

Checks and trust boundary. The recorded runtime is Python 3.12.14, python-flint 0.9.0 and FLINT 3.6.0. Separate rational checkers reconstruct the containment, sign, contraction and remainder inequalities from dyadic balls. Independent mpmath runs corroborate selected roots, integrals and coefficients, but do not replace interval enclosures. File hashes establish provenance, not mathematical validity. The analytic proofs and interval computations depend on the correctness of the stated bounds, producer code and arithmetic libraries; there is no proof-assistant verification or independent referee report. The original claim-to-source map remains in `manuscript_core/CLAIM_SOURCE_MAP.md`. Fresh producer replays, the equation review and reproduction commands for this revision are in `manuscript_review/README.md`. The recorded replay reproduces the mathematical endpoints of 7,010 stored balls; repeated occurrences are counted separately. It does not constitute an independent implementation of every enclosure.

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